information. The goal is to minimize total cost arising from price impact and price drift. Almgren and Chriss (1999, 2000) expanded on the idea of trade schedule optimization by incorporating a risk aversion parameter and set out to balance two conflicting terms (price impact and risk) based on the investor's risk appetite. Their proposed market impact formulation contains the right shape and market impact properties (e.g., convex shape with dollar value) but their objective function results in a path dependent stochastic process with a difficult and slow solution.

Malamut (2002, 2006) devised an approximated QP formulation and provided insight into parametric trade schedules to solve a non-linear impact formulation. Kissell, Glantz and Malamut (2004) incorporate a drift term into the objective functions and offer alternative goals to mean-variance optimization such as maximizing the probability of outperforming a specified cost (e.g., maximize the Sharpe ratio of the trade). Obizhaeva and Wang (2005) study an inter-temporal (not static) trade sequencing problem. They seek to solve a path dependent problem similar to Almgren and Chriss (2000) by understanding the half-life of a trade (e.g., the time for temporary impact to dissipate). Their techniques, however, are only presented for a single stock order.

In the financial literature, the mean-variance portfolio optimization of Markowitz (1952) clearly stands out as one of the more important quantitative approaches. The technique is widely used by portfolio managers and is an effective tool to manage risk and improve returns. However, mean-variance optimization is mostly used in the context of a one-period investment model. Li and Ng (2000) derive a solution to the multi-period mean-variance optimization problem where the allocation decision is reviewed in every period. Therefore, the proposed solution is dynamic since the decision to invest is reviewed after each period's results are known.

In this chapter, we present a multi-period trade schedule optimization approach for portfolio optimizers (Malamut, 2002). We offer four approaches that can be used to solve the trader's dilemma in an amount of time that can be useful for traders. These approaches expand on techniques presented in *Optimal Trading Strategies* (2003) and introduce real-time adaptation techniques to determine when it is appropriate to take advantage of market conditions given the overall risk composition of the trade basket.

TRADER'S DILEMMA

A typical trading situation is as follows: traders are provided with a basket of stock (e.g., program, trade list, portfolio, etc.) to transact in the market.